Table S1 Evaluation-set construction and activity provenance

A. Selection conditions

Target pair	Selection threshold	Chemical selection record	Deposited quota / composition	Eligible n	Selection replay status
EGFR/HER2	6	Murcko cap 5 within class	deposited N=110 with classes 28/38/32/12 before the later unresolved-arm exclusion	109	Documented; not replayed
JAK1/JAK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
JAK1/TYK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PIK3CA/mTOR	6	Fixed 110-molecule membership	32 / 32 / 32 / 14	110	Not independently replayed in this compilation
AChE/BChE	6.5 / 5.5	ChEMBL ID-prefix diversity key in deposited script	N=100; dual / A-only / B-only / neither 28/28/28/16	98	Documented; not replayed
F2/F10	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARG/PPARA	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARA/PPARD	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.

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B. Activity-source coverage

Target pair	n	Both ends traced	Complete identifiers	Missing identifiers	Source / endpoint mix	Threshold crossing (n)	Value / label mismatch
EGFR/HER2	109	109	101	8	22	13	0
JAK1/JAK2	110	110	109	1	9	0	0
JAK1/TYK2	110	110	108	2	13	4	0
PIK3CA/mTOR	110	110	109	1	24	3	0
AChE/BChE	98	98	98	0	8	2	0
F2/F10	110	110	110	0	10	0	0
PPARG/PPARA	110	110	110	0	42	9	0
PPARA/PPARD	110	110	110	0	40	4	0

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.